RESEARCH PAPER | DESCRIPTIVE LEARNER INTELLIGENCE

Learner Demographics and Course Enrollment Behaviour on EduPro

Who learns, what they choose, and why it matters

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7,516
enrollments

2,762
active learners

2.72
courses per learner

26-35
top age group

Includes: analysis of age, gender, category, type and level | 15 pages | 15 figures
Project: EduPro Learner Lens

DEMO BUILD - SYNTHETIC TEST DATA - NOT REAL EDUPRO RESULTS

SECTION 01 | OVERVIEW

Abstract and contents

This paper profiles the learners of EduPro, an online learning platform, and explains how their age and gender relate to the courses they choose. It joins the Users, Courses and Transactions sheets into one master table of **7,516 enrollments** made by **2,762 active learners** across **64 courses** and **7 categories**, recorded between 01 Jan 2023 and 30 Jun 2025. The work is descriptive: it answers who is learning and what they pick, not who will buy next.

The largest age group is **26-35**, which supplies 41.4% of all enrollments. Learners are 53.8% Male and 43.4% Female. **Programming** is the most enrolled category (23.5%), and Beginner courses form the biggest level group (44.0%). A learner takes 2.72 courses on average, and the most active 20% of learners account for 40% of enrollments. The paper closes with evidence-based recommendations for course design, marketing and inclusion.

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15	Recommendations and conclusion
	Pages are fixed, so the table of contents is exact. Figures 1 to 15 follow the same order as the sections.

How to read this paper

Percentages are shares of enrollments unless stated. A *preference index* compares a group's share of a category with the platform-wide share: 100 means the same as everyone, 150 means 50% more than average.

SECTION 02 | CONTEXT

Introduction and objectives

Online learning platforms serve people of different ages, genders, goals and subject interests. For a platform such as EduPro, knowing who the learners are is the base for designing relevant courses, improving engagement, targeting outreach and making the platform inclusive and accessible. EduPro holds user and transaction data, but until now decisions on course creation and marketing have rested on intuition rather than evidence.

Problem statement

- Which age groups are most active on the platform?
- How do enrollment patterns differ by gender?
- What course categories do different learner segments prefer?
- Are beginner, intermediate or advanced courses more popular within particular age groups?

Objectives

- Integrate the three data sheets with verified referential integrity.
- Measure the age and gender profile of active learners and their participation level.
- Describe enrollment by category, type and level, and name the most and least popular.
- Cross demographics with preferences (age by category, gender by level, age by level) to find learner segments.
- Quantify behaviour: courses per learner, concentration among active users, beginner versus advanced patterns.
- Turn the findings into practical recommendations for course creation, marketing and inclusion.

Scope and boundaries

The project is **descriptive learner intelligence**. It does not predict future enrollments, model price or revenue, or claim that age or gender *causes* a choice. Differences are reported as associations, with a significance test and an effect size so the reader can judge how large a difference really is.

Where each question is answered

Key analytical question	Section	Figures
What is the age distribution of learners?	5 Age profile	1, 2
How does enrollment vary across age groups?	5 and 8	2, 7
Are there gender-based differences in course selection?	6 and 9	3, 4, 8, 9
Which course categories attract the most enrollments?	7 Enrollment distribution	5, 6
Do beginners prefer certain course types or levels?	10 and 12	10, 14

> **Why descriptive analytics first**
> A clear picture of the present learner base is the foundation any later forecasting or personalisation depends on. If the baseline is wrong, a model built on top of it will be wrong in a more confident way.

SECTION 03 | METHOD

Data and methodology

The source is a single Excel workbook, *EduPro_Online_Platform.xlsx*, holding three linked sheets.

Sheet	Fields used	Role
Users	UserID, UserName, Age, Gender	Who the learner is
Courses	CourseID, CourseName, CourseCategory, CourseType, CourseLevel	What was offered
Transactions	TransactionID, UserID, CourseID, TransactionDate	Links learner to course; one row = one enrollment

Analytical pipeline

Stage	What was done	Output
1. Integration	Cleaned text, removed duplicate IDs, validated ages, joined Users-Transactions-Courses on UserID and CourseID	Master table, one row per enrollment
2. Demographics	Age bands <18, 18-25, 26-35, 36-45, 46+; gender distribution; participation per group	Age and gender tables
3. Distribution	Enrollment counts by category, type and level; Category Popularity Index	Rankings, shares
4. Cross-analysis	Age by category heatmap; gender by level; age by level; chi-square test and Cramér's V	Cross-tabs, preference index
5. Behaviour	Courses per learner, Lorenz-style concentration curve, Gini coefficient, level segments	Engagement metrics
6. Delivery	Five KPIs, Streamlit dashboard, this paper, executive summary	Deliverables

Definitions

- **Active learner**: a user with at least one enrollment in the merged data.
- **Enrollment**: one transaction row linking a learner to a course.
- **Category Popularity Index**: a category's enrollments divided by the average per category, times 100.
- **Preference index**: a group's share of a category divided by the platform share, times 100.
- **Gini coefficient**: 0 means every learner takes the same number of courses, 1 means one learner takes everything.
- **Cramér's V**: effect size for a cross-tab, 0 (no link) to 1 (perfect link).

> **Age bands**
> The project brief lists 36-45 and 45+, which overlap at 45. This paper uses **<18, 18-25, 26-35, 36-45 and 46+** so that every learner falls in exactly one band.

SECTION 04 | DATA

Data integration and quality

Joining the three sheets produced **7,516** merged enrollment rows. Each join was validated as many-to-one, so a transaction can never be duplicated by the merge. Referential integrity was checked in both directions before any analysis began.

Check	Result
Users rows read / duplicate UserIDs removed	3,000 / 0
Users with invalid or missing age	0
Users with missing gender (set to Not Specified)	0
Courses read / duplicate CourseIDs removed	64 / 0
Transactions read / duplicate TransactionIDs removed	7,516 / 0
Transactions with unparseable date	0
Transactions pointing to an unknown UserID	0
Transactions pointing to an unknown CourseID	0
Same learner enrolled in the same course more than once	0
Registered users with no enrollment (not active)	238
Final merged rows / unique learners / unique courses	7,516 / 2,762 / 64

Quality verdict
No broken keys, bad dates or invalid ages were found, so the merged table can be used as it stands.

Cleaning rules

- Text fields were trimmed and standardised (for example M and F became Male and Female).
- Ages outside 5 to 100 were treated as errors and left blank, never replaced by an average.
- A missing gender became *Not Specified*, so no learner silently disappears from totals.
- Of 3,000 registered users, 2,762 are active learners (92.1%). Demographic findings describe active learners only.

SECTION 05 | DEMOGRAPHICS

Age profile of learners

Active learners range from 15 to 57 years. The mean age is **28.9**, the median **28** and the standard deviation 8.5. The **26-35** band is the most active, with 41.3% of learners and 41.4% of enrollments. Together, 26-35 and 18-25 supply 67.4% of all enrollments, while the smallest band, 46+, supplies only 3.4%.

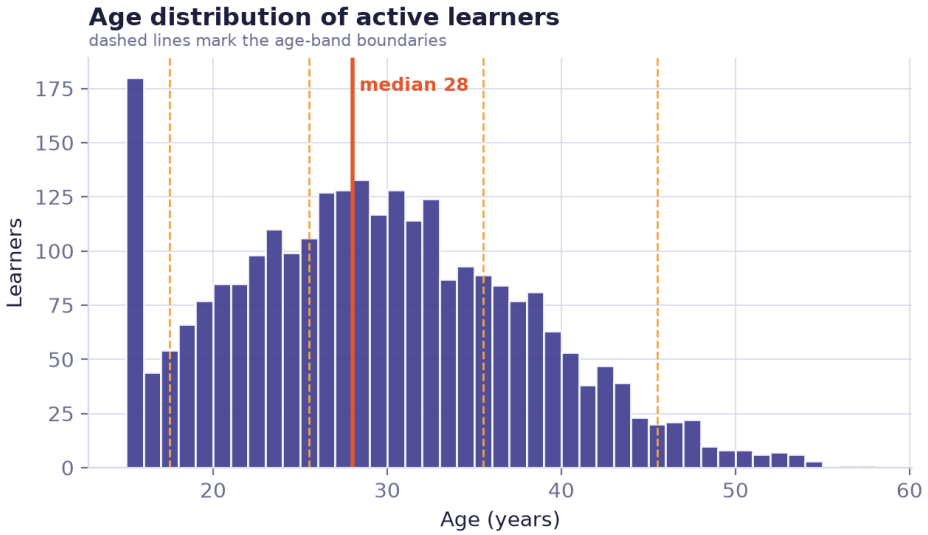


Figure 1. *Age distribution of active learners with age-band boundaries and the median.*

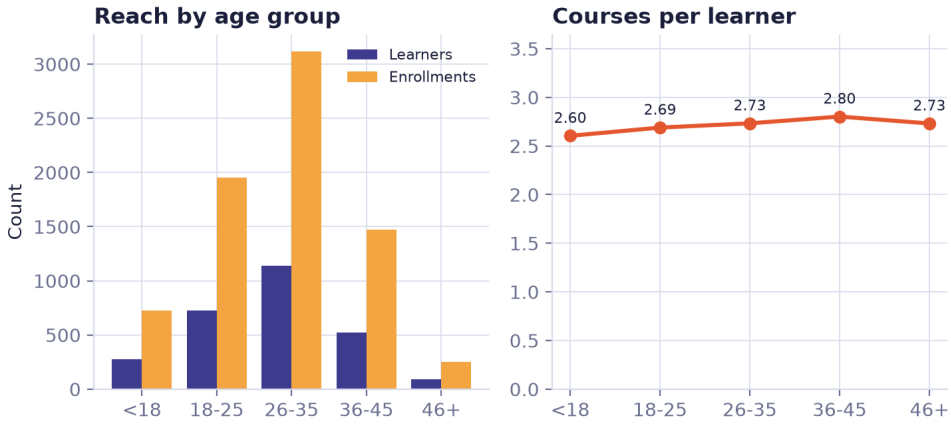


Figure 2. *Learners and enrollments per age group (left) and average courses per learner (right).*

Age group	Learners	Share	Enrollments	Share	Courses / learner
<18	278	10.1%	724	9.6%	2.60
18-25	726	26.3%	1,952	26.0%	2.69
26-35	1,140	41.3%	3,115	41.4%	2.73
36-45	525	19.0%	1,471	19.6%	2.80
46+	93	3.4%	254	3.4%	2.73

36-45 learners take the most courses on average (2.80), and <18 learners the fewest (2.60). Reach is therefore driven mostly by *how many* people join in each age band rather than by how hard they study.

SECTION 06 | DEMOGRAPHICS

Gender participation

Male learners are the largest group at 53.8%, followed by Female at 43.4%. The participation ratio of Male to Female is **1.24**, meaning there are 1.24 Male learners for every Female learner. Enrollment share closely follows learner share in every group (right panel), so gender differences in reach come from who joins, not from who enrolls more often once they have joined.

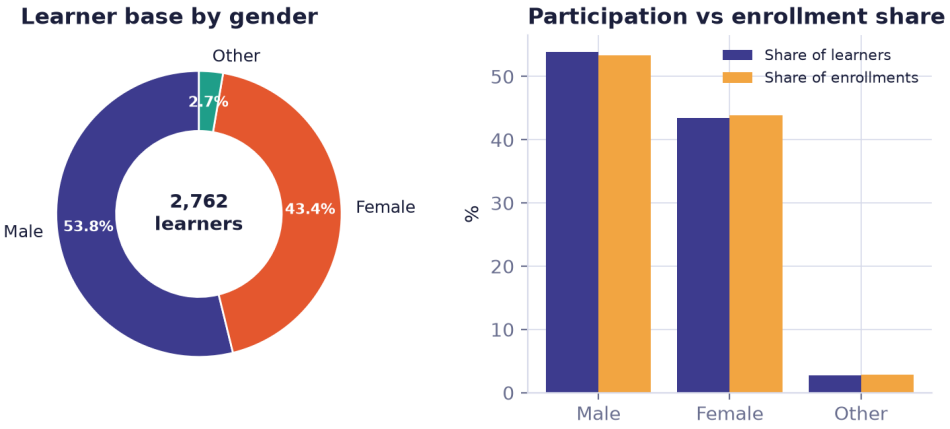


Figure 3. Learner base by gender (left); share of learners versus share of enrollments (right).

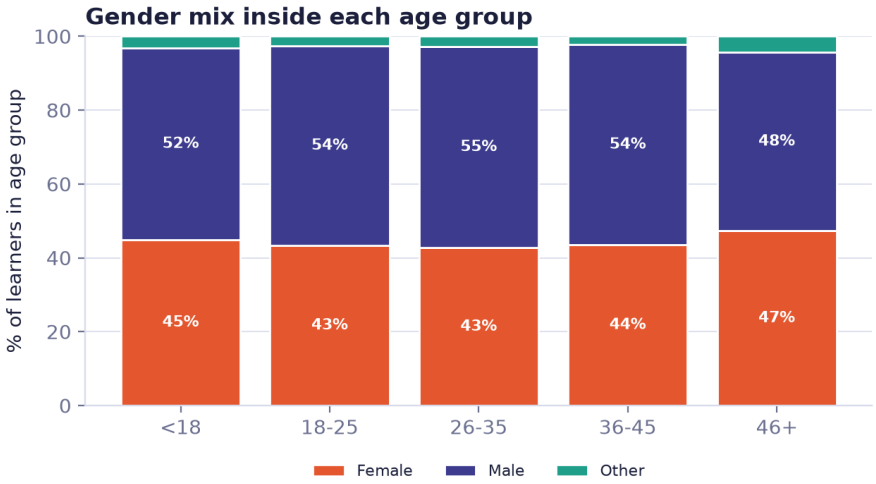


Figure 4. Gender mix within each age group. Bars sum to 100% per group.

Gender	Learners	Share	Enrollments	Share	Courses / learner
Male	1,487	53.8%	4,006	53.3%	2.69
Female	1,200	43.4%	3,294	43.8%	2.75
Other	75	2.7%	216	2.9%	2.88

Inclusivity read-out

Female learners make up the largest share of 46+ learners (47%) and the smallest share of 26-35 learners (43%), a spread of 5 percentage points. The gender mix is nearly the same across age bands, so a gender-balance programme can be run platform-wide rather than band by band.

SECTION 07 | ENROLLMENTS

Enrollment distribution

Programming is the most popular category with 23.5% of enrollments, and **Personal Development** is the least popular with 8.7%, a gap of 2.7 times. The top three categories (Programming, Data Science, Business) hold 58.5% of demand. The category with the most courses is Programming; the category that attracts the most enrollments *per course* is Programming (126), which shows where each new course is likely to be best used.

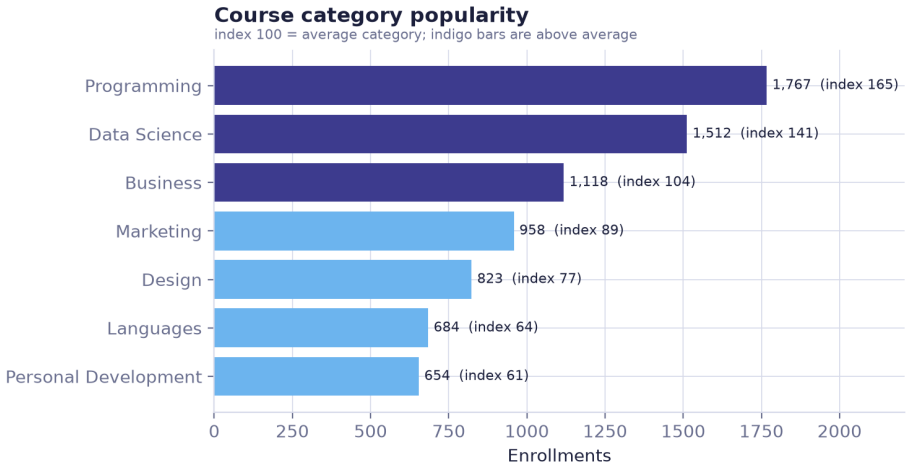


Figure 5. Enrollments by course category with the Category Popularity Index.

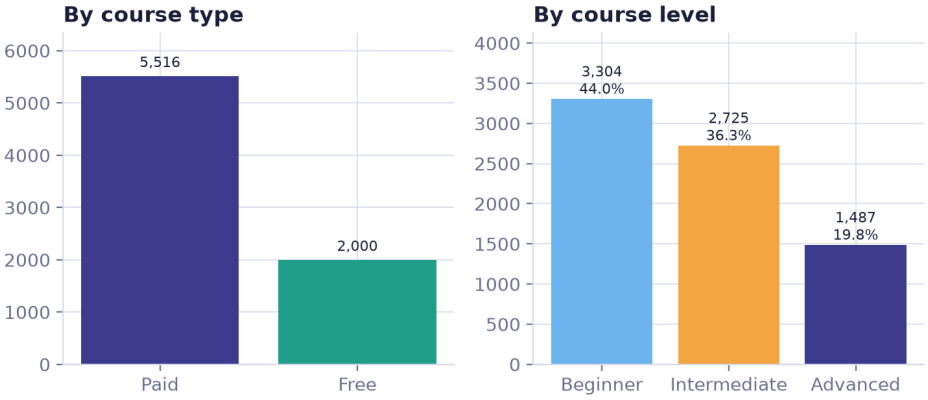


Figure 6. Enrollments by course type (left) and course level (right).

Category	Courses	Enrollments	Share	Popularity index	Per course
Programming	14	1,767	23.5%	165	126
Data Science	12	1,512	20.1%	141	126
Business	10	1,118	14.9%	104	112
Marketing	8	958	12.7%	89	120
Design	8	823	10.9%	77	103
Languages	6	684	9.1%	64	114
Personal Development	6	654	8.7%	61	109

By type, **Paid** courses take 73.4% of enrollments. By level, **Beginner** leads with 44.0% while Advanced is smallest at 19.8%.

SECTION 08 | DEMOGRAPHICS X PREFERENCE

Age and category preference

The heatmap on the left shows each age group's category mix; the one on the right converts it to a preference index so that groups of different size can be compared fairly. The strongest over-representation is **46+** learners in **Languages** (index 255, i.e. 155% above the platform share). The strongest under-representation is **46+** learners in **Programming** (index 40).

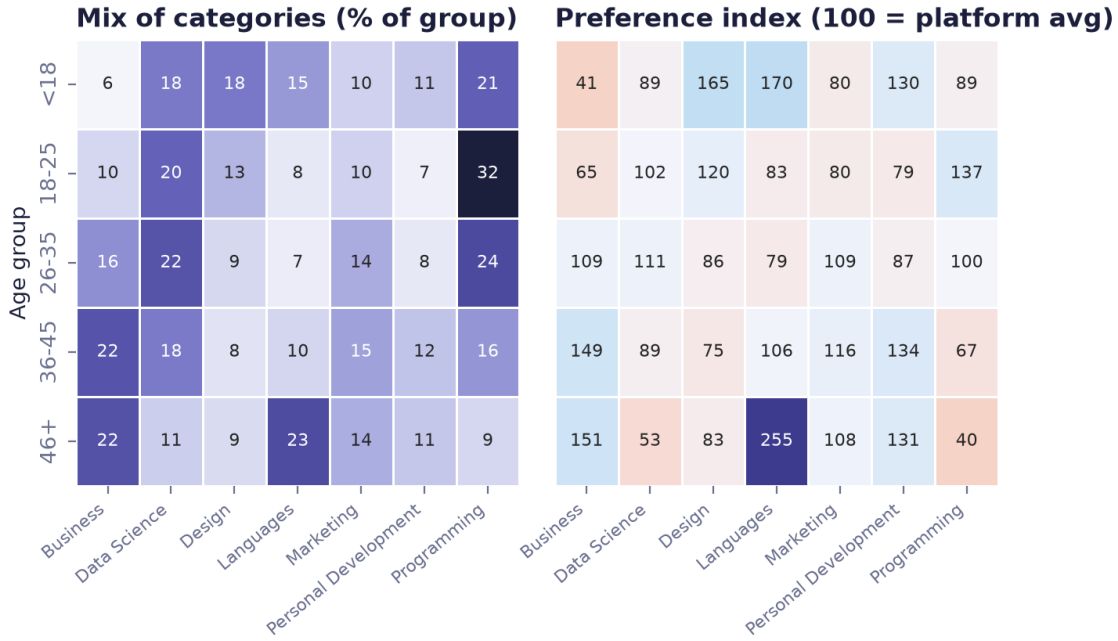


Figure 7. Age group by category: share of the group's enrollments (left) and preference index (right).

What each age band leans towards

Age group	Top category (share)	Most distinctive (index)
<18	Programming (21%)	Languages (170)
18-25	Programming (32%)	Programming (137)
26-35	Programming (24%)	Data Science (111)
36-45	Business (22%)	Business (149)
46+	Languages (23%)	Languages (255)

A chi-square test of independence on age group and category is statistically significant ($p < 0.001$), with a Cramér's V of 0.13, which indicates a **weak** association. The test shows whether the pattern is larger than chance; the effect size shows whether it is large enough to act on.

Practical meaning
A category that is popular overall can still be weak in one age band. Marketing messages and course bundles should follow the distinctive category of each band, not just the platform-wide leader.

SECTION 09 | DEMOGRAPHICS X PREFERENCE

Gender, level and type

Between Male and Female learners, the widest category gap is in **Data Science**, where Male learners place 1.8 percentage points more of their enrollments than Female learners; in **Languages** the gap runs the other way by 1.3 points. The widest level gap is in **Advanced** courses (0.7 points). A chi-square test of independence on gender and category is not statistically significant (p = 0.12), with a Cramér's V of 0.03, which indicates a **negligible** association.

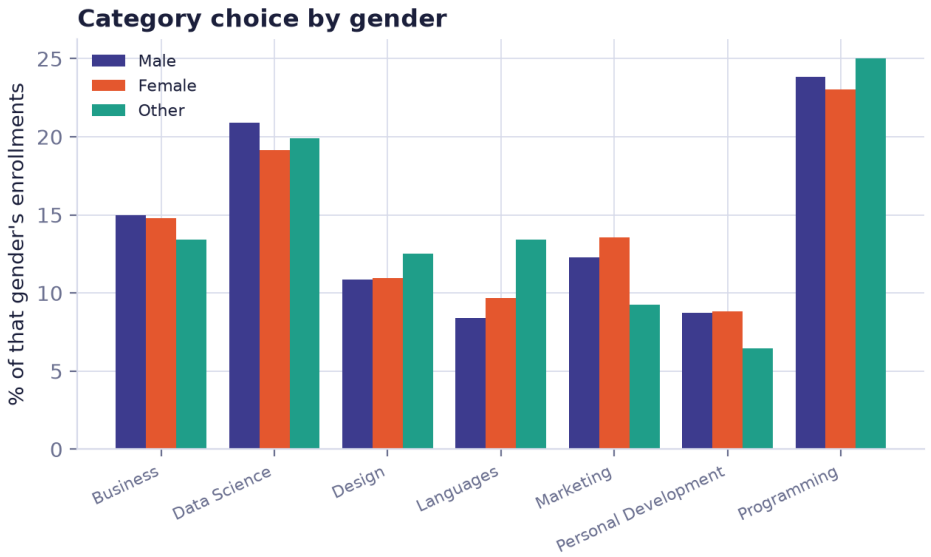


Figure 8. Category choice by gender, as a share of each gender's enrollments.

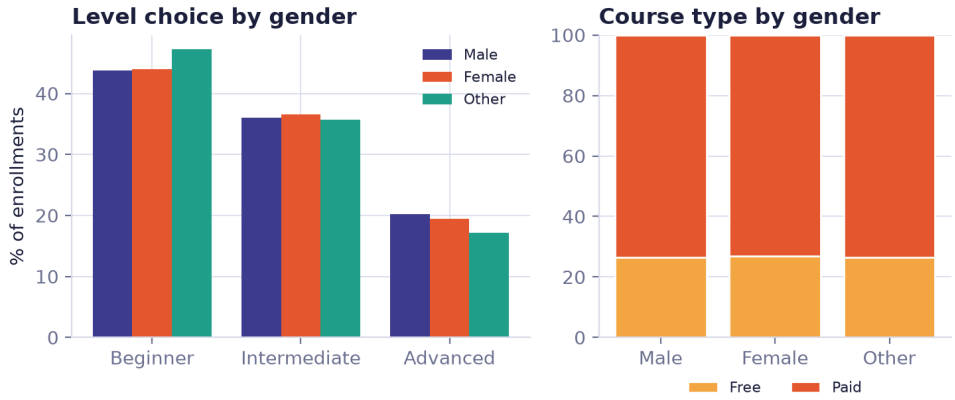


Figure 9. Course level (left) and course type (right) by gender.

Gender	Beginner	Intermediate	Advanced	Top category
Male	43.8%	36.0%	20.2%	Programming
Female	44.0%	36.6%	19.5%	Programming
Other	47.2%	35.6%	17.1%	Programming

A chi-square test of independence on gender and level is not statistically significant (p = 0.73), with a Cramér's V of 0.01, which indicates a **negligible** association. Where the effect size is weak, the safest reading is that men and women (and other learners) choose courses far more alike than different, and that differences in *reach* matter more than differences in *taste*.

SECTION 10 | DEMOGRAPHICS X PREFERENCE

Age and course level

46+ learners lean most on Beginner courses (47% of their enrollments), while **18-25** learners lean least (42%). Advanced courses are taken most by **46+** learners (25%) and least by **18-25** learners (19%). A chi-square test of independence on age group and level is statistically significant (p = 0.021), with a Cramér's V of 0.03, which indicates a **negligible** association.

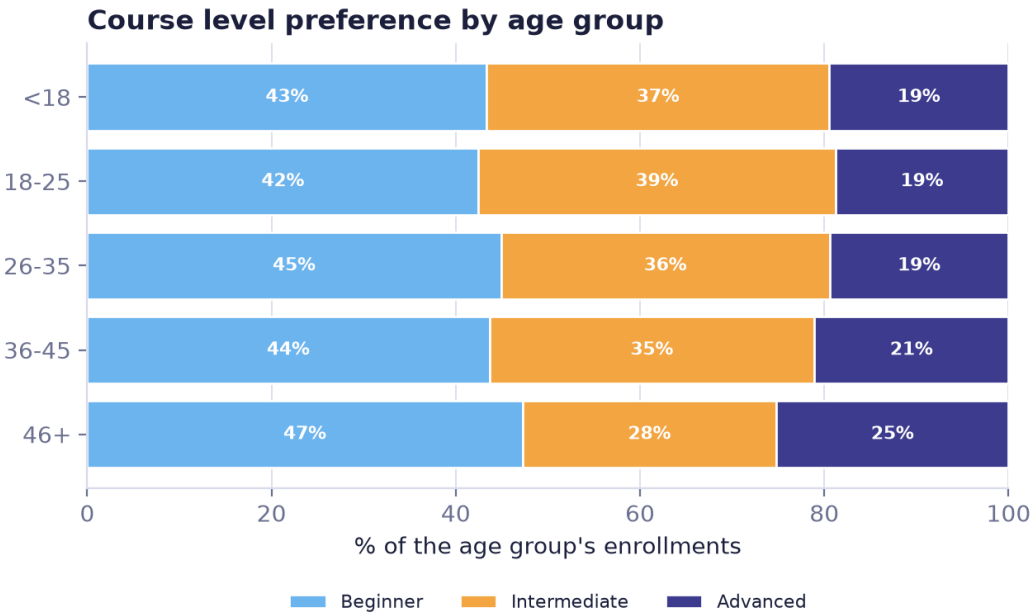


Figure 10. Level mix of each age group's enrollments.

Age group	Beginner	Intermediate	Advanced	Preferred level
<18	43.4%	37.2%	19.5%	Beginner
18-25	42.5%	38.7%	18.8%	Beginner
26-35	44.9%	35.7%	19.4%	Beginner
36-45	43.6%	35.3%	21.1%	Beginner
46+	47.2%	27.6%	25.2%	Beginner

Reading the pattern

If the Advanced share rises or falls steadily with age, it suggests that skill depth is linked to life stage and that learning paths should be designed per band. If it stays flat, level choice is better explained by prior experience than by age, and placement tests or self-assessment would be a better targeting tool than age.

Course type by age group

Age group	Free	Paid
<18	25.8%	74.2%
18-25	23.1%	76.9%
26-35	28.6%	71.4%
36-45	27.3%	72.7%
46+	28.0%	72.0%

SECTION 11 | BEHAVIOUR

Learner behaviour and concentration

A learner takes **2.72** courses on average (median 2, maximum 8). 30% of learners have taken only one course, and 27% have taken four or more. On average a learner has explored 2.23 categories, and 66% have crossed into more than one category.

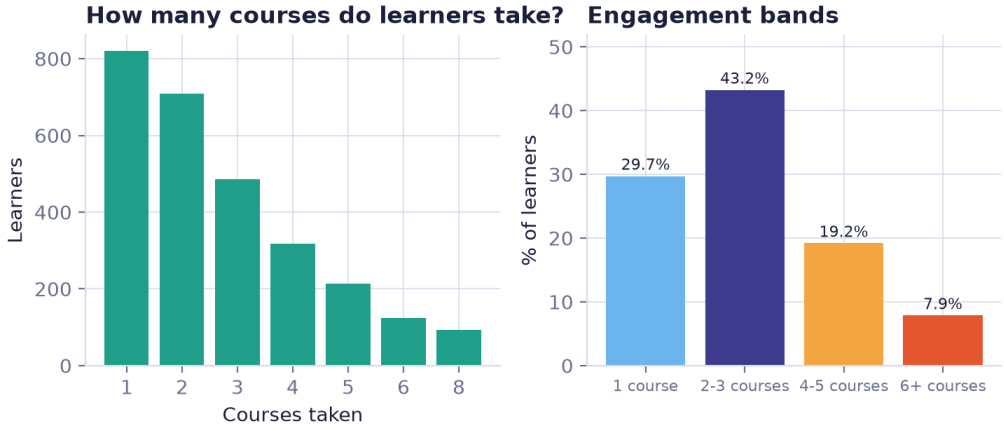


Figure 11. Distribution of courses per learner (left) and engagement bands (right).

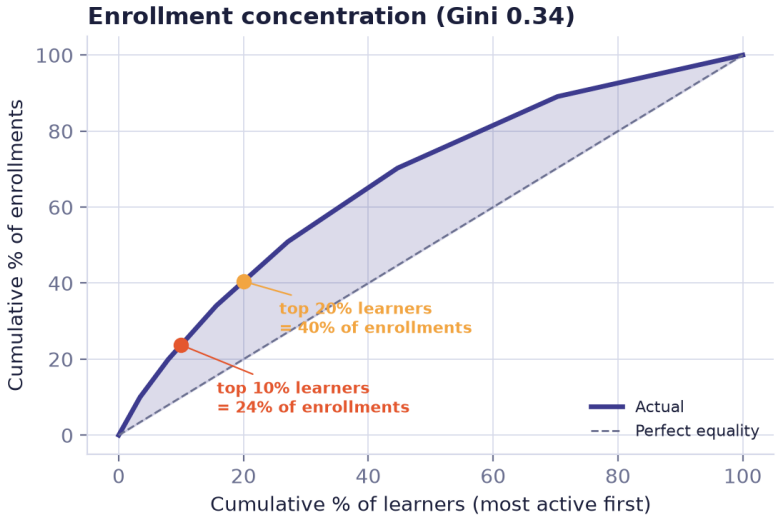


Figure 12. Concentration curve: how much of total enrollment the most active learners generate.

Group	Avg courses	Median	Max
Age <18	2.60	2	8
Age 18-25	2.69	2	8
Age 26-35	2.73	2	8
Age 36-45	2.80	2	8
Age 46+	2.73	2	8
Gender Male	2.69	2	8
Gender Female	2.75	2	8
Gender Other	2.88	2	8

Enrollment is moderately concentrated. The most active 10% of learners generate **24%** of enrollments and the most active 20% generate **40%**. The Gini coefficient is **0.34**. Retaining the most active learners protects a disproportionate share of demand, and moving single-course learners to a second course is the clearest route to growth.

SECTION 12 | BEHAVIOUR

Skill progression and timing

To compare beginner and advanced learners, each learner is placed in a segment by the *highest* level they have taken.

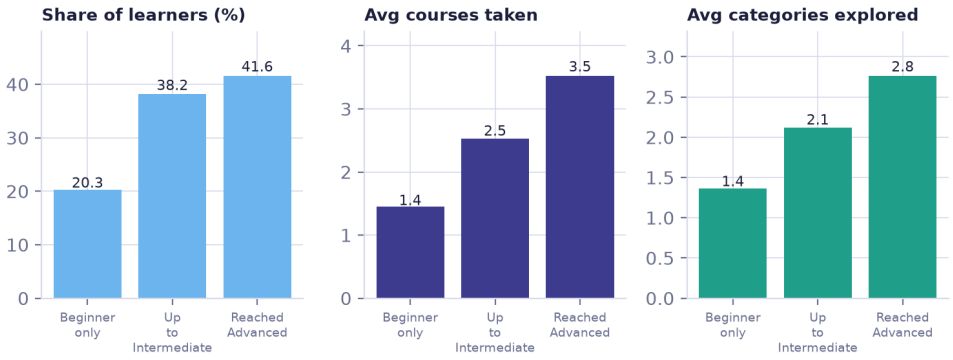


Figure 13. Learner segments by highest level reached: share, average courses, average categories.

Segment	Learners	Share	Avg courses	Avg categories	Avg age
Beginner only	560	20.3%	1.45	1.36	28.9
Up to Intermediate	1,054	38.2%	2.53	2.12	28.2
Reached Advanced	1,148	41.6%	3.52	2.76	29.5

Learners who have reached the top level take 3.52 courses on average, against 1.45 for beginner-only learners. Part of any gap is mechanical, because a learner who takes more courses has more chances to meet an advanced one, but it still shows that advanced learners are the platform's heaviest users.

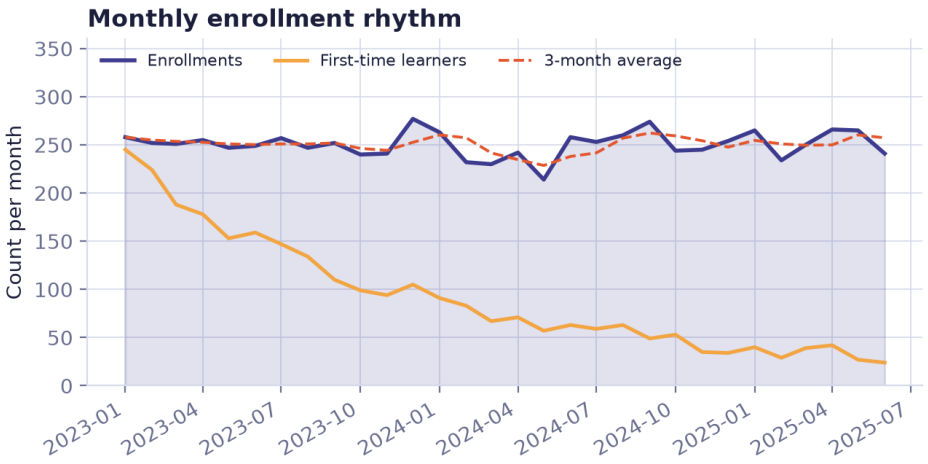


Figure 14. Monthly enrollments, first-time learners and a 3-month moving average, 01 Jan 2023 to 30 Jun 2025.

Demand peaked in **December 2023** (277 enrollments) and was lowest in May 2024 (214); the monthly average is 251. Comparing the second half of the period with the first, monthly enrollments stayed broadly flat. Sunday is the busiest weekday and Thursday the quietest.

SECTION 13 | SYNTHESIS

Learner segments and KPI scorecard



Figure 15. Headline indicators for the platform.

The five required KPIs

KPI	Value	Reading
Total Enrollments	7,516	Platform engagement indicator
Enrollments by Age Group	Top: 26-35 (41.4%)	Demographic reach (full split in Section 5)
Gender Participation Ratio	Male:Female = 1.24	Inclusivity metric
Category Popularity Index	Programming (165)	Course demand; 100 = average category
Level Preference Distribution	Beginner (44.0%)	Skill maturity of the learner base

Learner segments and what they prefer

Segment	Learners	Top category	Most distinctive (index)	Preferred level
Age <18	278	Programming (21%)	Languages (170)	Beginner (43%)
Age 18-25	726	Programming (32%)	Programming (137)	Beginner (42%)
Age 26-35	1,140	Programming (24%)	Data Science (111)	Beginner (45%)
Age 36-45	525	Business (22%)	Business (149)	Beginner (44%)
Age 46+	93	Languages (23%)	Languages (255)	Beginner (47%)
Gender Male	1,487	Programming (24%)	Data Science (104)	Beginner (44%)
Gender Female	1,200	Programming (23%)	Languages (106)	Beginner (44%)
Gender Other	75	Programming (25%)	Languages (148)	Beginner (47%)

SECTION 14 | ACTION

Recommendations and conclusion

Recommendations

- **1. Protect the core category.** Programming draws 23.5% of enrollments. Keep it fresh with new courses and update the most-enrolled titles before adding breadth elsewhere.
- **2. Add courses where each course works hardest.** Programming earns 126 enrollments per course, the best in the catalogue. New course slots there are the lowest-risk investment.
- **3. Rebuild or reposition Personal Development.** At 8.7% of enrollments it is the weakest category. Test whether the gap is a content-quality problem or a visibility problem before retiring it.
- **4. Market by band.** The 26-35 band is the biggest (1,140 learners) and leans to Data Science. Use each band's most distinctive category in its campaigns, for example Languages for 46+ learners.
- **5. Grow the thinnest band deliberately.** 46+ learners are the smallest group. Partnerships, scholarships or format changes aimed at this band widen reach without competing with the core.
- **6. Close gender gaps with targeted outreach.** Gender differences are mainly in reach (Male 54%), not in course taste (Cramér's V 0.03). Focus on the age bands where the smaller gender is least represented (Section 6).
- **7. Build pathways for single-course learners.** 30% of learners took one course. Bundles, next-course prompts and level-to-level pathways convert them into repeat learners.
- **8. Reward and retain the active core.** The top 20% of learners drive 40% of enrollments. Loyalty offers and early access protect that base.
- **9. Match level mix to audience.** Offer more Advanced content where that band is under-served (18-25: 19%) and more Beginner on-ramps where that share is low.
- **10. Monitor quarterly.** Re-run this pipeline each quarter (one command per step) and track the five KPIs over time.

Limitations

- The analysis describes active learners and enrollments only; completion, ratings and revenue are outside the data.
- Associations are not causes. A chi-square test shows a pattern is unlikely to be chance, not why it exists.
- Age is a snapshot per learner, and small age bands can give unstable percentages; counts are always shown beside shares.

Conclusion

EduPro's learner base is centred on the 26-35 age band and tilted towards Male learners; Programming leads demand and Beginner courses lead by level. The platform's most useful next moves are to target messages by age band, add courses where each course performs best, and turn single-course learners into repeat learners.